



Aberdeen *Group*

Enterprise Service Bus: an SOA Middleware Foundation

Next Steps in SOA Series

June 2006

— Sponsored by —

Fiorano and IBM: see sponsor_content directory

↓ DO NOT DELETE SECTION BREAK BELOW THIS LINE ↓



Executive Summary

An enterprise service bus (ESB) is messaging middleware software that provides the secure interoperability and message transport services between application “services” in a Service Oriented Architecture (SOA) computing environment via XML, Web services interfaces, messaging adopters, and rules-based routing. In researching ESB adoption, Aberdeen’s hypothesized that since an ESB provides the SOA “highway” infrastructure, ESB adoption is a proxy for overall market adoption of SOA technology. This report details the findings and analysis of our ESB benchmark survey, but also contains considerable new insights into the maturation of other SOA services at mid-2006.

Key Business Value Findings

Users are splitting into three technology camps, which we label *SOA Lite*, *SOA ERP*, and *SOA Enterprise*. *SOA Lite* is for users who are primarily deploying web services that do not require mission-critical capabilities such as high-volume scalability, high availability and failover, management, governance and security. *Enterprise SOA* requires and uses mission-critical SOA middleware capabilities. *SOA ERP* is used by companies who are choosing to deploy SOA surrounding their Enterprise Resource Planning (ERP) application software.

The overall adoption rate of SOA technology is very robust, with 90% of survey respondents saying they will exit 2006 with experience in SOA planning, design, and/or programming under their belts.

Company size plays a large role in differentiating SOA technology experience; one-fifth of enterprise-size organizations have more than a year’s experience with SOA programming but only one-sixteenth of mid-size organizations. Almost 80% of small companies have no immediate plans to start SOA programming.

SOA Lite

SOA Lite categorization is justified by the fact that nearly half say they have no plans to start using an ESB, while only one-ninth have more than a year’s experience using ESB software. This category, which is dominated by small-to-mid-size companies and Microsoft .Net sites, is predominantly focused on light-weight Web services.

Enterprise SOA

Enterprise SOA today is synonymous with large enterprises of over \$1 billion. These companies have been using SOA technology for longer than a year, have complex integration issues, and have the most to gain from the benefits of SOA, and typically set high standards for uptime and recoverability. Enterprise SOA incorporates the greatest functionality in adapting to cross-platforms, integration flexibility (especially with legacy EAI investments), ESB suite and SOA suite functionality, and mission-critical volume and uptime capabilities.

There are technology challenges to *Enterprise SOA*: half report concern over scaling to production volumes, reliability, and scalability; creating a metadata repository/registry for ESB data elements; and establishing ESB security, governance and management. The



services and organizational strategies to overcome the technology challenges are to slow the pace of mission-critical adoption, beef up internal training and technology teaming, and to use outside services in a “learn while doing” fashion. Complicating the learning curve is that fact that while half are implementing SOA using a Java-based J2EE environment, about one-third are implementing on both Java and Microsoft .Net.

SOA ERP

One-sixth of respondents view SAP as a primary SOA vendor. (Oracle does slightly better but sells its SOA middleware standalone as well as with its applications). We conclude that a significant minority of companies — especially mid-size organizations — will use the SOA integration capabilities built into the latest versions of ERP vendor’s application products in two fashions: as the preferred integration way *into* the applications; and, as a low-cost SOA toolkit for integration from the ERP applications *out* to other applications and processes. Virtually every application software company that Aberdeen follows has an SOA enablement development program delivered or underway.

Implications & Analysis

SOA Adoption

SOA is a major technology wave that is driven by the unbearable costs of technology integration, which is programmer labor-intensive. By making it easier to integrate line-of-business-mandated process changes, IT is embracing SOA as a long-term cost-reducer and tool for time-to-market.

In comparison with previous information technology adoption cycles, SOA’s rush to early adoption is breath-taking. SOA in general and ESB specifically are not dramatically changing IT software buying patterns. Figure 1 shows primary SOA vendor and primary ESB choices for installed products:

- First, the incumbent Enterprise Application Integration (EAI) vendors BEA Systems and webMethods are the leader and number three, respectively. Half the respondents and two-thirds of BIC are considering EAI side-by-side with an ESB purchase.
- Second, Oracle and SAP have a combined 19% “primary vendor” designation, an indication that *SOA ERP* is a valid category.
- Third, .Net makes Microsoft the primary vendor only for those who use .Net alone, not for the one-third who use Java and .Net.
- Fourth, IBM has the highest primary vendor recognition in pure SOA technology and in ESBs, while Fiorano and Iona are the leading choices in pure-play ESBs.

ESB Deployments Match a Robust SOA Roll-Out

Using Aberdeen’s PACE framework, the overall survey group is driven to deploy ESBs and SOA technology by management of IT complexity and the need to speed IT implementations, while Best-in-Class (BIC) organizations are driven by the needs to align with the business and to re-use applications via Web services.



ESB adoption is a large enterprise (>\$ 1 billion) phenomena, with about 7% saying they have no plans to start using an ESB in their SOA roll-out. However, the majority of mid-size and almost four-fifths of small companies have no plans to use an ESB, yet they are already designing, and programming SOA applications. These data indicate smaller companies are adopting the *standards* of SOA and the related programming, but are not yet ready to adopt an SOA-style of development for mission-critical applications — or they will rely on *SOA ERP* to close the technology gap.

Three-fifths are creating composite applications that support application, with over one-third planning customer-facing B2C applications connected to back-office applications.

An eye-opening 46% will deliver more than three major application integration projects using SOA in 2006.

Over three-quarters of the ESBs are deployed at companies that describe themselves as *centralized* by geography, business unit, IT, or corporate-wide.

Overall, respondents report plans to deploy between 3,700-5,500 servers for edge applications and from 2,700-4,500 servers at datacenters. Thus, ESB servers in our survey alone are driving about 0.15 to 0.25% of the 4 million servers we project 2006. The implication is that servers running ESB — and the rest of the SOA stack — are beginning to impact enterprise server hardware purchases. We believe SOA will drive significant hardware purchases since the SOA value equation essentially trades off computing inefficiency (e.g., XML) for higher labor productivity. At this point in the history of computing, trading hardware and SOA software for programmer labor is the right approach for both short- and long-term value generation.

ESBs were considered alongside EAI, business process management (BPM), and IBM Message Queue or CORBA message brokers, in that order. Buying evaluations are driven by ease of integration flexibility with current and planned applications; ESB business process control, change, management, government, and life-cycle capabilities; and the completeness of the ESB product's offering (commonly called an ESB suite).

The key factors in buying decisions are overall technical attractiveness, ROI/ROA projections, and installation services. Slightly less than half plan to standardize on SOA software from a single vendor. About the same percentage will employ a best-of-breed strategy with ESB and other SOA software purchases, believing industry standards for SOA will minimize integration risks. We believe both approaches have merits.

Recommendations for Action

SOA is a long-term technology journey, and consequential decisions must be made along the way. We are at one such juncture. Companies must consider and decide whether they will conscientiously head down the *SOA Lite*, *SOA Enterprise*, or *SOA ERP* roads. Backtracking later will add costs and delays.

SOA middleware is often deployed as a suite consisting of application server, ESB, repository/registry, security, and management/governance. We believe IT practitioners are making the right choice by investing in SOA suites, primarily because it will be costly to go back and retrofit already developed applications with SOA products such as security and governance at a future date.



We concur with the decision by respondents to preserve their existing EAI middleware investments while moving forward with SOA technology, rather than a rip-and-replace of EAI middleware.

↓ DO NOT DELETE THE SECTION BREAK BELOW THIS PARAGRAPH ↓

Table of Contents

Executive Summary	i
Key Business Value Findings	i
Implications & Analysis	ii
Recommendations for Action.....	iii
<i>Chapter One: Issue at Hand</i>	1
Application Integration is a Huge Challenge to Agility	2
<i>Chapter Two: Key Business Value Findings</i>	4
How Can We Take Enterprise Applications to the Next Level?	5
<i>Chapter Three: Implications & Analysis</i>	10
Process and Organization	11
Technical Maturity and Technology Usage	11
Integrating Applications with SOA Technology.....	9
<i>Chapter Four: Recommendations for Action</i>	10
Laggard Steps to Success	19
Industry Norm Steps to Success	19
Best in Class Next Steps	20
Featured Sponsors	21
Sponsor Directory	22
Author Profile	23
Area of Focus	23
Current Research	23
Experience	23
<i>Appendix A: Research Methodology</i>	24
<i>Appendix B: Related Aberdeen Research & Tools</i>	26
About AberdeenGroup	27



Figures

Figure 1: Field Service Practices Are a Prime Determinant of Business Performance.....	Error! Bookmark not defined.
Figure 2: Pressures that companies are feeling are as follows.....	3
Figure 3: Consumer-Driven Firms Lead Strategic Field Service Push.....	4
Figure 4: Proactively Reaps Rewards.....	9
Figure 5: Metrics Tied to Success.....	9
Figure 6: Mobility Provides Boost.....	11
Figure 7: Utilities Firms Focus on Field Service Improvement.....	19

Tables

Table 1: Field Service Challenges and Responses.....	5
Table 2: Field Service Competitive Framework.....	10
Table 3: Field Service Technology Investments in Next 12-24 Months	10
Table 4: PACE (Pressures, Actions, Capabilities, Enablers)	12
Table 5: PACE Framework.....	Error! Bookmark not defined.
Table 6: Relationship between PACE and Competitive Framework.....	Error! Bookmark not defined.
Table 7: Competitive Framework.....	Error! Bookmark not defined.

Chapter One: Issue at Hand

Key Takeaways

- Existing application integration technology is too complex, resource-consuming, and slow-to-implement to keep up with business process changes.
- SOA technology from both application ISVs and development/middleware companies is the preferred technology base for solving the application integration problem.
- ESB is not the first of many successive SOA products to be deployed. It is often deployed with a suite of SOA middleware products.

SOA Solves IT Integration Cost and Complexity Problems

There are significant benefits in successfully implementing application integration in support of business process change. The differences in application software maintenance costs, which include process-based IT changes, are over 11% of the IT budget between best-in-class and average organizations. Given \$1.3 trillion in global enterprise IT spending in 2006, the business value on the table is an eye-popping \$143 billion a year, so a modest and achievable one-sixth improvement by average and laggard enterprises towards the best-in-class results in \$23.4 billion a year in business value.

In late 2005 and into 2006, Aberdeen studies explored many challenges that enterprises face in keeping up with, integrating, and paying for the complex changes that are constantly needed to keep business processes efficient and competitive. For example, in *Achieving More Value from Enterprise Applications*, between 40% and 50% of respondents said that redesigning business processes, high related integration costs, and customization-related IT challenges were major application integration stumbling blocks.

The data in Figure 1 also typify the responses we received in research documented in *SOA in the Supply Chain*, *SOA in IT*, and *The Product Quality Benchmark* of manufacturers. It is statistically proven that the cost and agility needed to keep up with business process changes driven by the line-of-business is a C-level opportunity for improvement. Based on the huge 11% of IT budget gap between process- and technology-disciplined Best-in-Class organizations and Industry Norm organizations, the issue should be high on the executive agenda.

Getting there will be an opportunity for dramatic improvement or it could be another technology swim in the swamp as has happened so often in the past. There are two major starting points for most enterprises: many business processes are spread across several application “silos”, making integration tough to begin with; and, prior technology such as enterprise application integration (EAI) software got

Competitive Framework Key

The Aberdeen Competitive Framework defines enterprises as falling into one of the three following levels of practices and performance:

Laggards (30%) — practices that are significantly behind the average of the industry

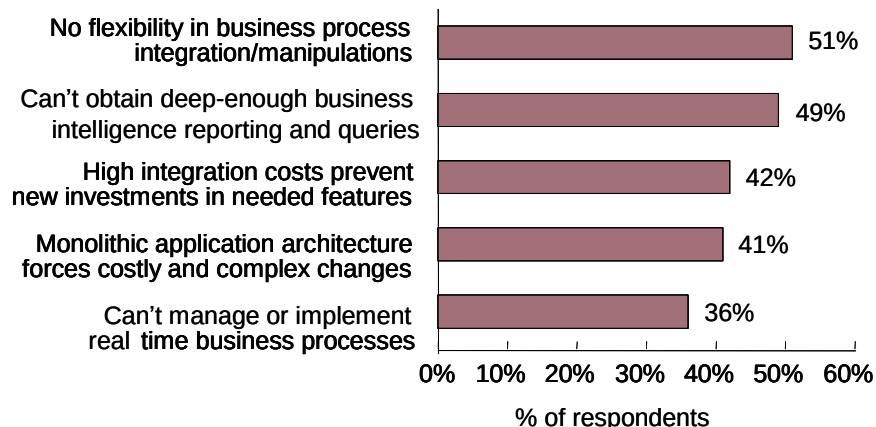
Industry norm (50%) — practices that represent the average or norm

Best in class (20%) — practices that are the best currently being employed and significantly superior to the industry norm



the job done once but has proven very expensive to maintain in terms of programmer labor.

Figure 1: Application Integration Stumbling Blocks

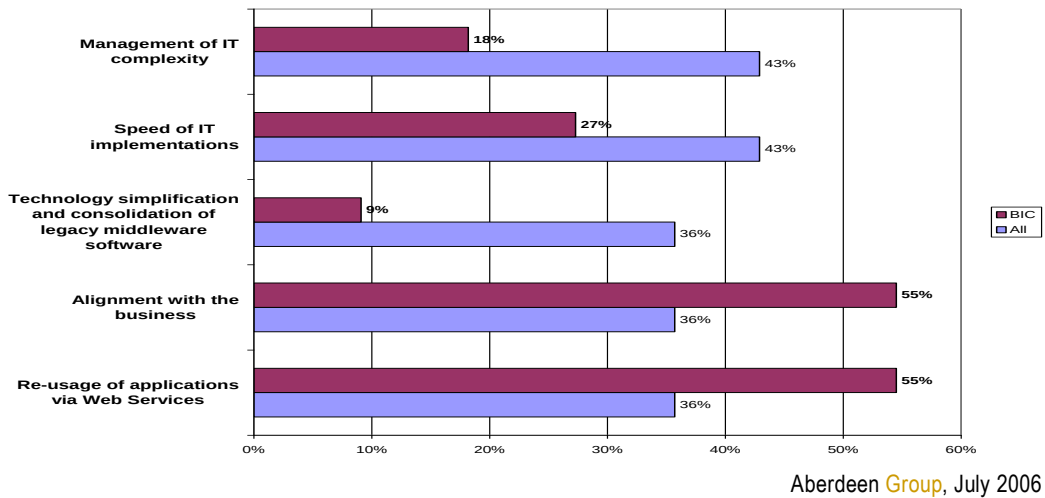


'Enabling More Value from Enterprise Applications' [Aberdeen Group](#), May 2006

ESB technology is supplementing EAI and related integration technologies with communications connectors, business-process orchestration and management, and life-cycle management capabilities that are proving to lower the complexity and costs of initial integration as well as ongoing changes because “reusability” of integration software investments is finally being widely achieved by SOA adopters.

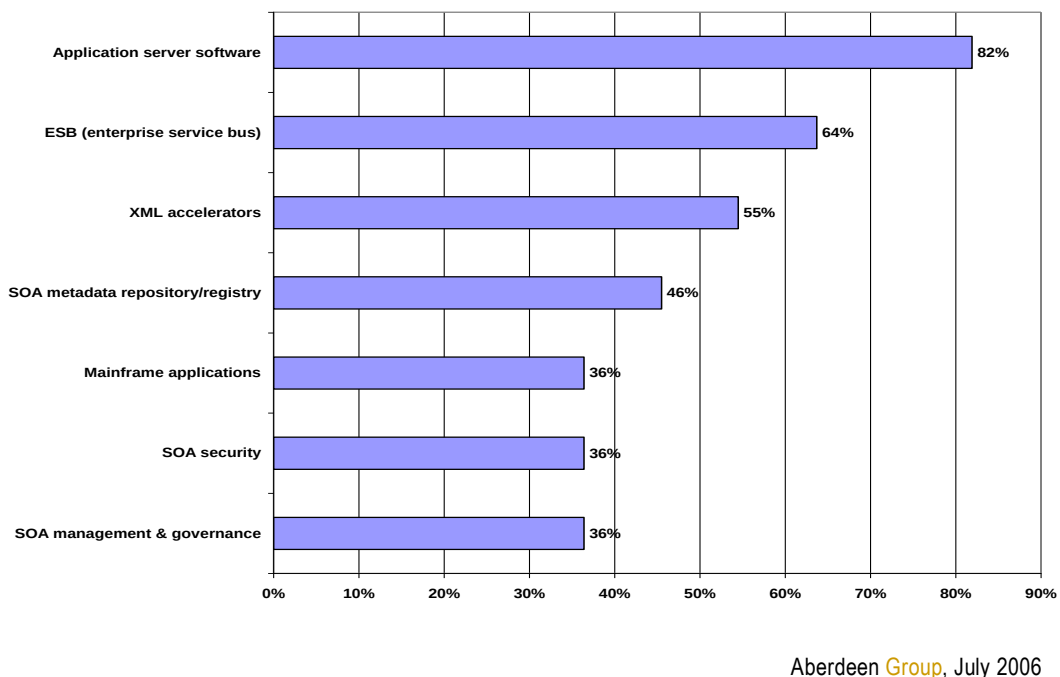
The SOA Middleware Market is Already Disparate

Figure 2 shows the factors driving adoption of ESB and SOA middleware technology. This graph illustrates the disparate nature of the SOA market, even though the vast majority of companies have less than a year's experience with ESB technology. Overall, the top drivers are managing IT complexity and speeding up the delivery of IT applications and changes. However, the 20% of respondents rated Best-in-Class in Aberdeen's competitive framework are focused on aligning with the business and re-using existing applications as Web services .

Figure 2: ESB Adoption Drivers (Multiple Choice)

ESB Products are Often Installed in an SOA Suite

Figure 3 illustrates the percentage of survey participants who had deployed specific SOA technologies. While about two-thirds have ESB software deployed, half have a metadata repository/registry also deployed and more than a third have security, management and governance products deployed as well. Telephone interviews indicate those who are deploying multiple products at once are consciously trading off an initial steeper learning curve for lower integration costs later.

Figure 3: SOA Middleware Presently Deployed



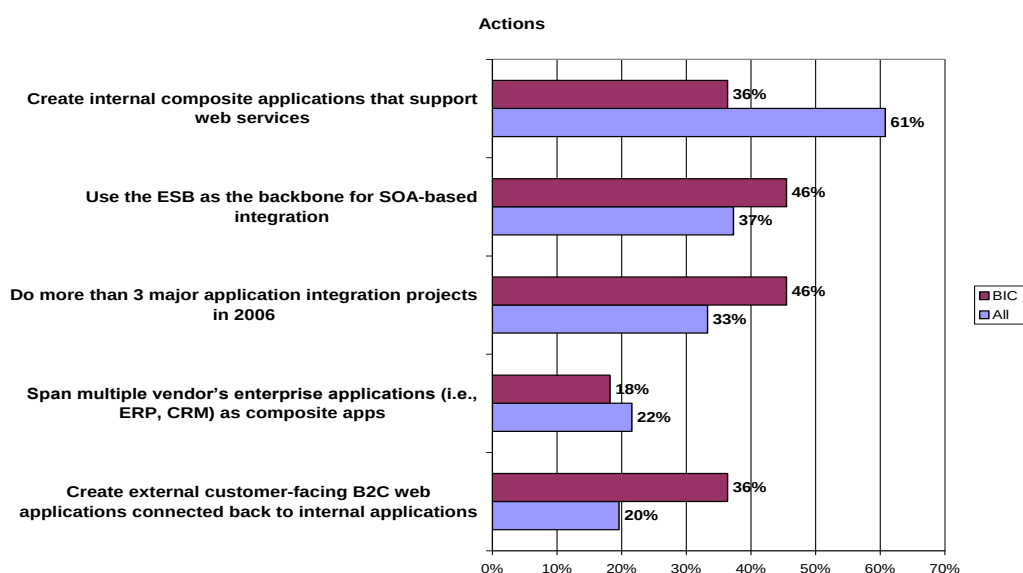
Chapter Two: Key Business Value Findings

Key Takeaways

- Over 90% are rapidly scaling the SOA adoption curve in 2006.
- ESB technology is complementing, not replacing, EAI technology.
- Enterprise ESB issues related to integration with registry/repository and scaling to high volumes are the greatest challenges faced by ESB practitioners.

Technicians are rapidly absorbing the technology behind SOA applications development and moving into deployment. By the end of 2006, the 90% of organizations that intend to adopt SOA will be at the programming stage.

Figure 3: Actions Undertaken (Multiple Choice)



Aberdeen Group, July 2006

Figure 3 depicts the actions undertaken in response to the drivers shown in Figure 2. The overall population is rushing to create composite applications that better mimic the needs of business processes, which often span applications. BIC organizations are using ESBs to create an “SOA highway” around the organization.

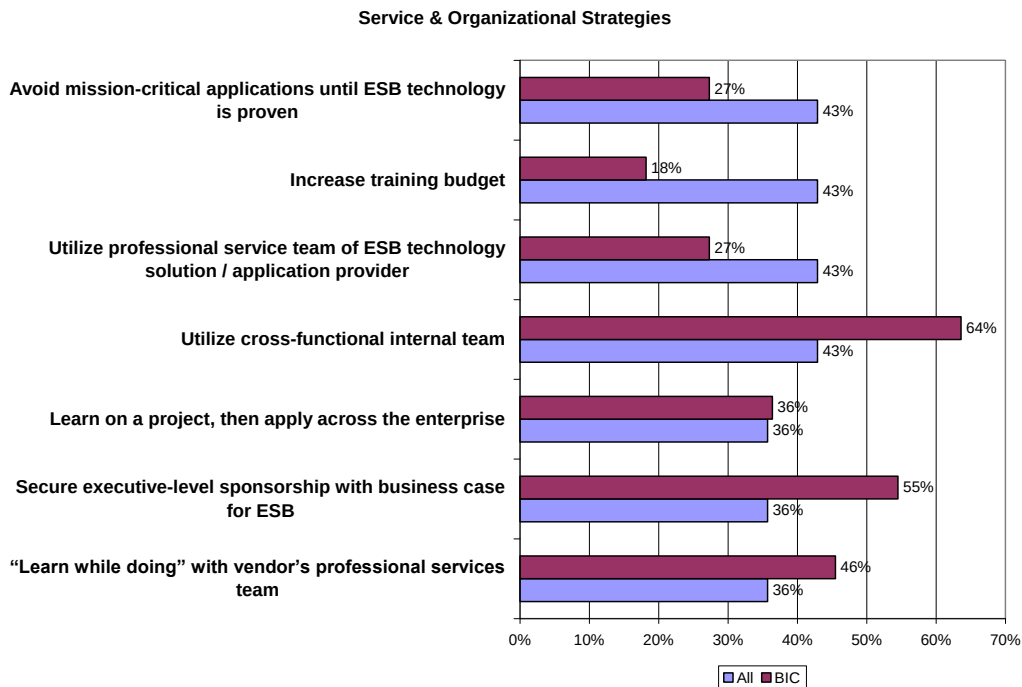
Importantly, the adoption of SOA is hardly cautious: half the BIC and a third of the overall sample in Figure 3 plan on undertaking four or more major application integration projects in 2006. The implications are that the SOA technology curve is not particularly

difficult to surmount; the backlog of SOA-enhanced integration projects is long; and that SOA development across the enterprise is in full swing.

Organizational and Services Strategies

The most common approach to development and deployment is to use a cross-functional team of SOA architects, systems programmers, programmers, and business process analysts (Figure 4). This approach is used by almost two-thirds of BIC organizations. The BIC are also “learning while doing” by working beside more expert outside services consultants and technicians. The overall sample is also looking to buy outside services and to increase the internal training budget. Last, the companies most focused on *Enterprise SOA* are proceeding cautiously to production, looking to ensure that mission-critical applications will meet Service Level Agreements on uptime and response times under load.

Figure 4: Organizational and Services Strategies



Aberdeen Group, July 2006

Coping With Multiple Frameworks and Platforms

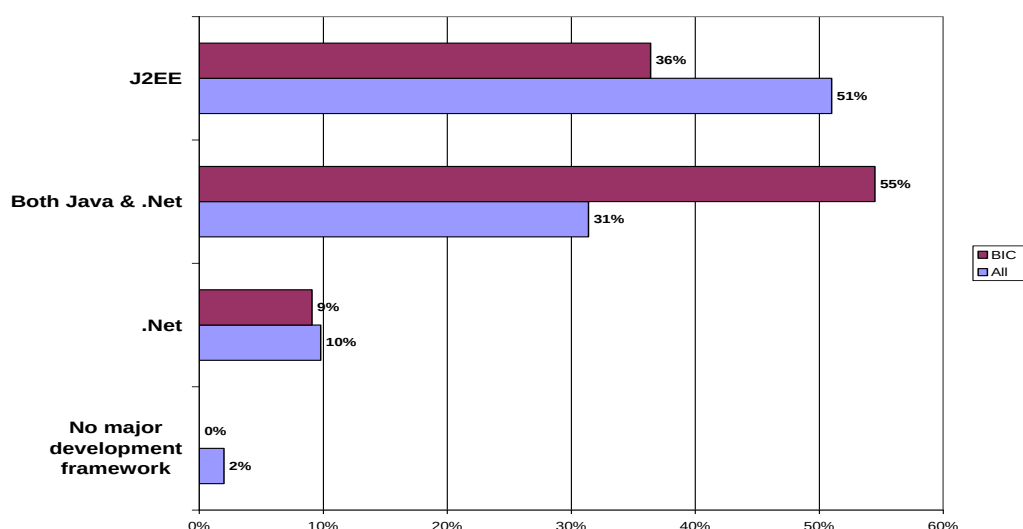
Over half the overall survey group and two-thirds of the BIC group are using Java as the target framework for their SOA development (Figure 5), compared to only ten percent dedicated to Microsoft’s .Net framework. Importantly, over half the BIC and a third of the overall group are planning deployments across *both* Java and .Net frameworks. However, it requires careful selection of programming tools to develop for both frameworks with a common development environment.



Java-only sites are planning deployments to more servers than .Net sites, by about 2 to 1. Reflecting the importance of SOA to large enterprises, almost two-fifths plan SOA application involving their mainframes. Overall, our survey identified deployment plans for at least 3,700 to 5,500 servers on the Edge and 2,700 to 4,600 servers at data centers. Clearly, SOA plays a significant integration role on the Edge.

Over 40% will employ a best-of-breed strategy with both ESB and SOA products, believing the strong adoption of industry standards within SOA technology will make best-of-breed integration risk low. One-third will standardize on a single ESB product and deploy widely.

Figure 5: SOA Target Frameworks



Source: [AberdeenGroup](#), April 2006

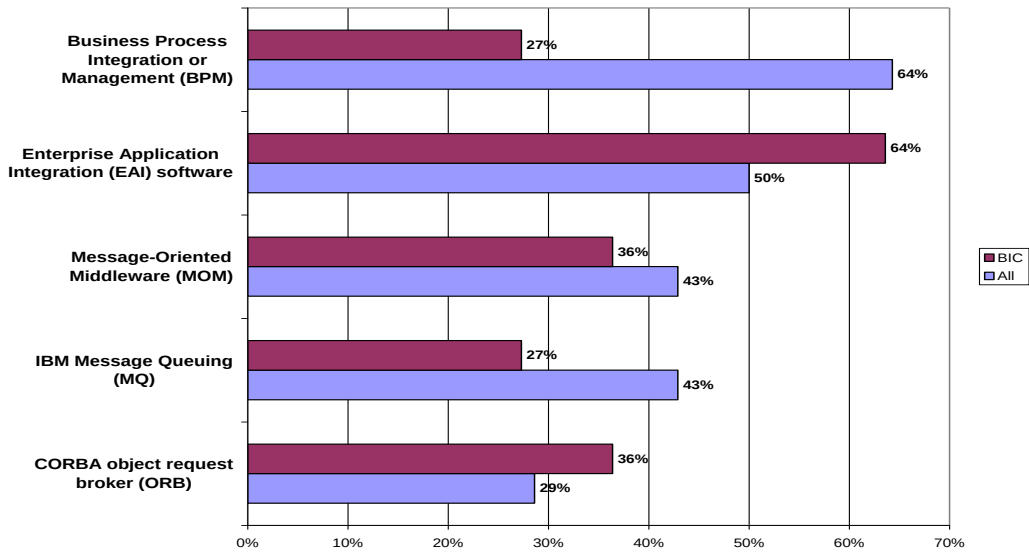
ESB Complements and Competition

ESB technology is being ignored by nearly half the survey, setting those companies towards an *SOA Lite* direction (Figure 6). Of the half that chose an ESB, two-thirds of the overall survey considered a business process integration or business process process management (BPM) software product. Many of today's ESB's have strong business process orchestration capabilities built in, so the comparison is valid.

Two-thirds of BIC organizations and half the overall survey chose an ESB over EAI products, while less than half looked at message-oriented middleware, IBM's message queuing (MQ) product, or the standards-based CORBA object request broker (ORB).

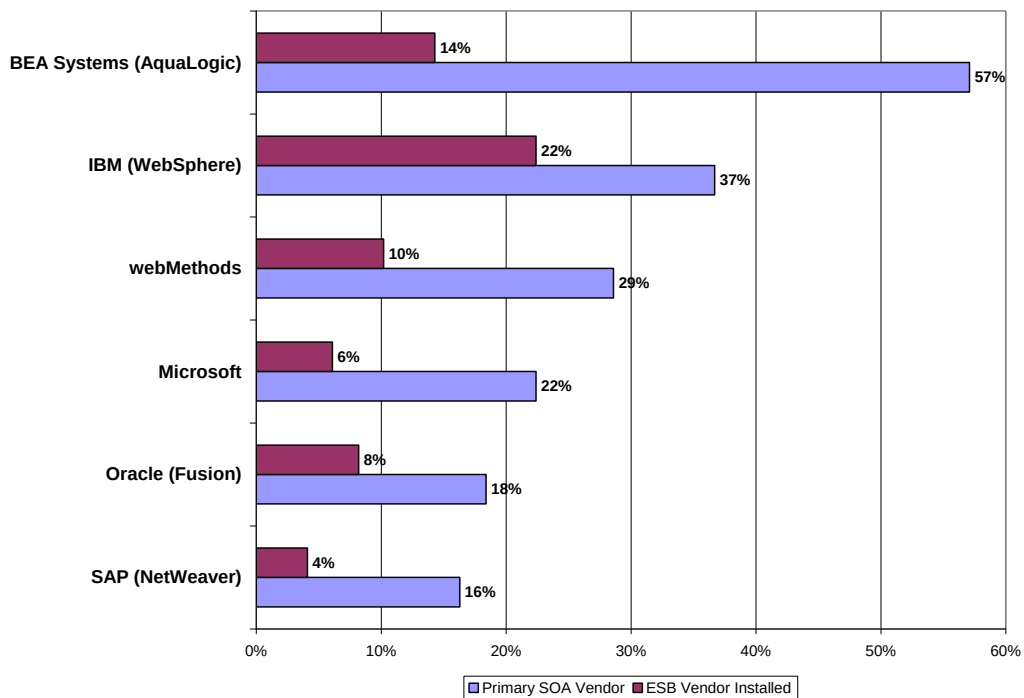
However, the picture is not all about competition. Quite the opposite, as illustrated in Figure 7. BEA Systems and webMethods, two EAI companies, sandwich IBM for the top three choices for primary SOA vendor. webMethods, for example, does not offer an ESB per se, but its products perform the functions of an ESB. "Functional equivalency" by trusted EAI suppliers is protecting the EAI investments at many companies, especially large enterprises.

Figure 6: Other Technologies Considered in ESB Selection



Source: [AberdeenGroup](#), April 2006

Figure 7: Primary ESB and Primary SOA Vendor (Multiple Choice)



Source: [AberdeenGroup](#), April 2006

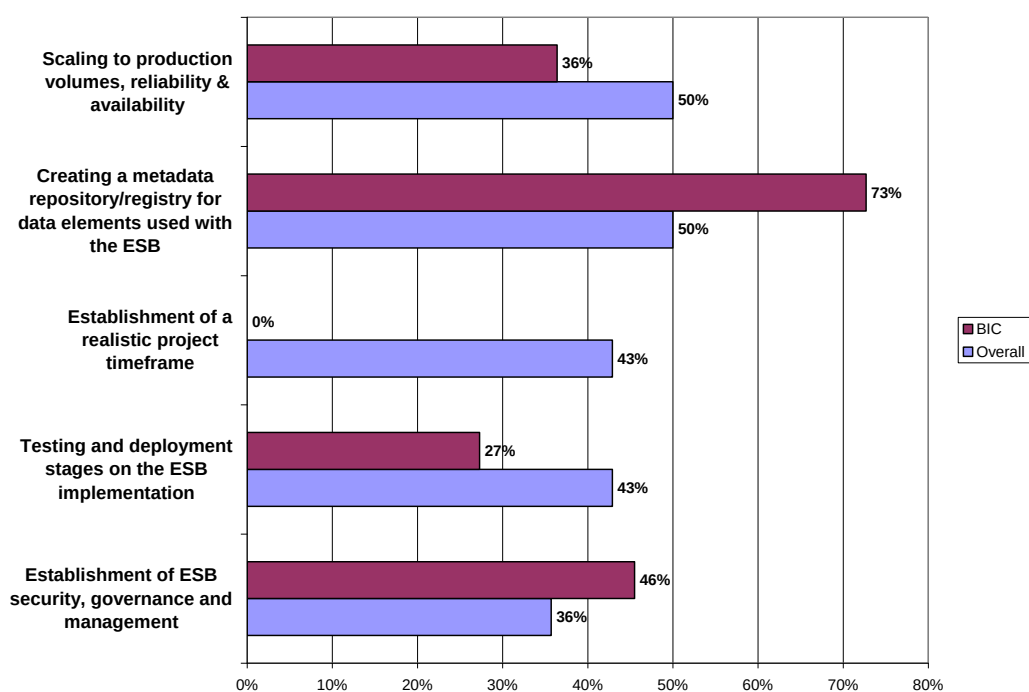


Getting to ESB Deployment Has Challenges

The vast majority of companies implementing an ESB are above \$500 million in revenue. Figure 8 illustrates the major challenges respondents encountered. Creating a metadata repository or (UDDI) registry stymied half the survey — and almost three-quarters of the BIC. A common mistake is underestimating the time and effort required to populate the repository/registry with the service rules needed by the ESB for regulating services. Half the overall population and a third of the BIC had difficulty scaling to production volumes and with service-level issues such as availability and reliability. Underestimating the hardware resources required to scale out a medium to large SOA implementation is a common challenge — and somewhat understandable given the lack of experience by most practitioners.

Other challenges were encountered by less than half of the survey, but are worth discussing. None of the BIC but two-fifths of the overall survey report difficulty in establishing a realistic project schedule. This comes from inexperience and is not a long-term problem. Testing and deployment is also problematic for a significant minority due to inexperience, the complexity of the service-registry-ESB interaction, and a dearth of testing tools. Lastly, almost half the BIC and a third of the survey overall report challenges in establishing security, governance, and management of an ESB. Security and management can be solved with SOA point products. Governance involves organizational issues related to ESB control, especially in the large number of distributed computing applications planned, and should be addressed at the design stage.

Figure 8: ESB Technology Stumbling Blocks (Multiple Choice)



Source: AberdeenGroup July 2006

Conclusions on the Business Value of ESB

Large companies are clearly voting with their wallets to buy and deploy ESBs. They are the most likely to have more than a year's experience in programming with SOA technology.

By focusing on creating composites of multiple software applications, practitioners are dealing with a major issue outlined in our study *Achieving More Value with Enterprise Applications*: today's business processes span multiple software application silos.

The aggressive plans of a significant minority to tackle four or more major SOA applications is indicative of the fact that practitioners are confident in their ability to execute, have a clear plan of attack, and are moving aggressively to deploy SOA-enabled applications to drive value in their organizations.

The qualitative data we gathered on SOA performance metrics during this study indicate organizations are reducing application maintenance costs as a percentage of the IT budget and increasing the percent of the IT budget allocation to line-of-business innovation — 5% would be an initial planning goal. Notably, those two metrics weigh heavily in Aberdeen's calculations of which organizations are Best-in-Class. Thus, we now have qualitative data that suggest SOA is delivering measurable business value to the IT department (and we have no doubt that SOA applications are delivering value to the line-of-business units).



Chapter Three: Implications & Analysis

Key Takeaways

- Best-in-class IT organizations have the right ISV software for functionality, at the current release, with a strong process for managing customizations by ROI.
- Best-in-class IT organizations have overcome the significant problems of integration.
- Best-in-class IT organizations have in-house competency to perform LOB integration requests and keep the infrastructure up to date.
- In best-in-class IT organizations, IT process maturity leads to lower integration and maintenance costs and frees up considerable capital for LOB innovation.

As shown in Table 1, survey respondents fell into one of three segments – Laggard, Industry Average, or Best in Class based on Aberdeen’s model for IT organizations that weights the cost of application maintenance as a percentage of IT budget and the percent of the IT budget that spent on delivering innovations to the LOB.

The three segments are further analyzed based on their characteristics in four key categories:

- (1) process (application maturity and experience, success with application integration, discipline in measuring business value);
- (2) organization (corporate focus/philosophy, level of collaboration among stakeholders, IT organization’s focus and the level of centralization);
- (3) technical maturity (competency in performing application integration, ability to complete integration in-house); and
- (4) technology (application modernity, functional adequacy, discipline in maintenance).

Table 1: SOA Adoption Competitive Framework

	Laggards	Industry Average	Best in Class
Process	Recent adopter of modern applications. Little experience in integration. ROI seldom measured.	Looks at SOA as cost-cutter, not process-improver. ROI planned before project.	Considerable experience and integration process is successful. Management by KPMs and SLAs.
Organization	Business focus on inventory management, revenue growth, and process standardization. Highly centralized.	Focus on order fulfillment, CRM, real-time visibility. Beginning to implement SOA for composite applications.	IT is decentralized and especially LOB-focused. Service-oriented. SOA viewed as new process enabler.

	Laggards	Industry Average	Best in Class
Technical Maturity	No plans for SOA. Low competency leads to high costs, delays, delivery inconsistency. Needs outside help.	Modest competency. SOA plans but no ESB plans. High maintenance costs. May need outside help. Under 12 months SOA programming.	More than 12 months SOA programming experience. Work done internally or side-by-side with outside experts. SOA architect in place.
Technology	Applications antiquated and inadequate. Customizations are poorly documented. No ESB or SOA infrastructure plans.	Project-oriented point purchases, SOA Lite or SOA ERP. Low-volume, low-complexity SOA like Web services. No ESB or other SOA middleware.	Well-documented and managed. Full SOA suite deployed. Anticipating scaling and availability issues.

Source: [AberdeenGroup](#), July 2006

Process and Organization

- In the process category, BIC firms that have low maintenance costs as a percentage of IT budget also show high innovation rates and considerable discipline in IT process management. Many are already quite experienced with SOA technology, and SOA business value benefits for lower IT application development and maintenance costs and faster IT delivery are showing up in performance metrics. Laggards are twice as likely to be focused on IT integration costs as the BIC, while laggards are one-fifth as likely to be mature enough to be focused on IT operating costs as the BIC.
- Organizationally, the BIC IT shop is organized and staffed with specialists and generalists who are brought together into multi-disciplinary teams for SOA projects. Outside services are engaged where necessary, but internal staff work along-side the contractors in order to “learn by doing” skills transfer. BIC are disciplined and focused on continually responding to LOB changes and optimizations. Laggards focus on unsophisticated business metrics such as revenue growth. They tend to be highly centralized, but least able to cope with change. For example, they seldom measure business value before or after implementing a project.

Technical Maturity and Technology Usage

- BIC have superior IT processes and talent, enabling them to show no significant challenge in managing the application integration process or the technologies involved.
- Average and Laggard organizations are most likely to adopt SOA Lite and SOA ERP. Average organizations have a propensity to be a version or more behind the ISV’s latest release. Since SOA and other new technologies are most likely in the latest releases, average and laggard organizations will be unable to leverage the flexibility of SOA technology – a widening gap between SOA-enabled haves and have nots is a point of technology differentiation.
- The majority of Average and Laggard enterprises have no plans to adopt ESBs or other components of an SOA suite in 2006. They may see the wisdom of



adopting SOA middleware after deploying more applications or when confronted by the integration limitations of Web services without the support of SOA middleware.

- BIC are most likely to be challenged by complex SOA implementation issues such as scaling to production volumes and achieving high-availability across a distributed SOA infrastructure. Clearly, many BIC organizations have taken the technology further and pushed it. Average and laggards will have to anticipate the lessons learned.
- BIC are more likely to use IBM WebSphere and/or EAI products from BEA Systems or webMethods. Average organizations are more likely to choose SOA ERP from the likes of Oracle or SAP, or to choose a best-of-breed ESB.
- Over 80% of BIC will buy an ESB based on ROI across multiple projects; BIC are buying SOA infrastructure with a goal of cross-company deployment while Average and Laggard are still focused on individual project-based ROI.

ESB Purchase Factors

ESBs are being purchased primarily by large enterprises and BIC IT organizations. Table 2 shows the factors important to an ESB buying decision as chosen by the majority of respondents, and where we are overweighting Very Important over Important.

The number one weighted factor is the ease of integration flexibility with current and planned applications. Respondents told us that since large enterprises have considerable investments in EAI technology, they cannot afford to rip-and-replace EAI with SOA; SOA must co-exist with EAI technology.

The number two factor is completeness of the ESB “suite”. This factor indicates buyers do not want light-weight ESB technology, but are looking for integrated capabilities for BPM, service management and governance, and service life-cycle features such as change management. The high rating of this factor tells Aberdeen that the largely large-enterprise buyers are selecting enterprise-class ESB products, rather than less capable products that might suffice for a single application project but could not be later scaled up to enterprise-class levels. This is sophisticated buyer behavior fairly early in the adoption cycle.

The remaining factors are all clustered closely together and may safely be viewed as a group. Many of these can be tied to the first and second place factors: ease of integration/EAI and a robust ESB suite.

What ESB buyers are *not* interested in is open source ESB technology.

Table 2 – Factors in ESB Purchase Decision

Weighted importance of specific drivers for the ESB technology decision			
	Very Important	Somewhat Important	Not Important
Ease of integration flexibility with current and planned applications	43%	25%	0%
ESB business process control, change, management, governance, and life-cycle features	29%	35%	6%

Completeness of vendor's SOA product offering	20%	43%	4%
Completeness of vendor's ESB product offering (e.g., suite)	20%	43%	4%
ESB security features	25%	37%	6%
ESB features protect our organization's legacy middleware investments	29%	31%	8%
ESB scalability, robustness, reliability, clustering, and fail-over features	25%	35%	4%
ESB process modeling and BPEL capabilities	20%	39%	8%
Extensive range of ESB communications connectors and transport options	22%	35%	12%
ESB business process orchestration capabilities	22%	35%	8%
ESB compliance with industry standards	12%	47%	6%
Proven ability of ESB to sustain high volumes in production	20%	35%	12%
ESB mediation capabilities	18%	37%	8%
ESB development environment	20%	33%	10%
ESB is tightly integrated with vendor's other SOA technology	22%	29%	10%
ESB is tightly integrated with vendor's ERP applications	14%	35%	12%

Source: [AberdeenGroup](#), July 2006

Challenges and Solutions Differ by Company Size

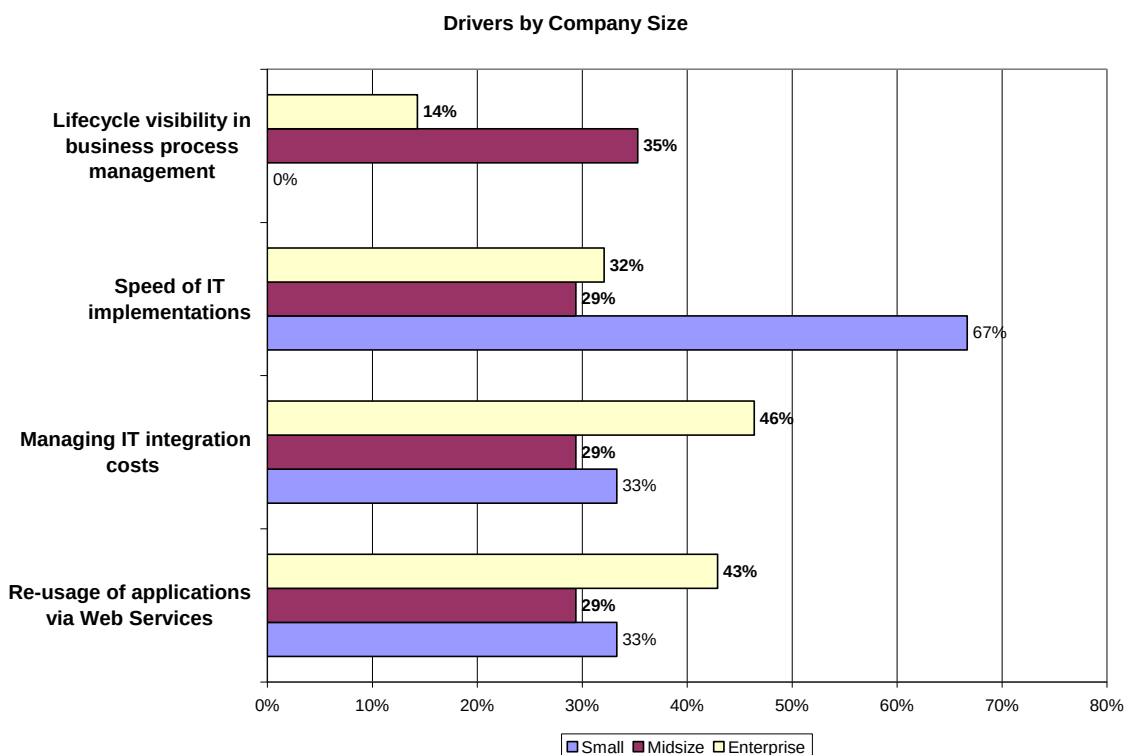
Companies are overcoming challenges with responses that differ by company size as measured by revenue (Figure 9) with responses (Figure 10).

Small Companies with less than US\$50 Million in Revenue

One-third of small companies are starting their SOA journey at the planning and design stage in 2006, and almost 30% will not have reached the programming stage this year. For whatever reason, three-quarters of small companies have no plans to buy and ESB or related SOA middleware. Small companies are fixated on the speed of IT implementations. When buying an ESB, small companies look first for completeness of the SOA suite, and next for tight integration of the ESB into the overall SOA suite. More than larger companies, those under \$50 million will look to the ESB as the backbone for ESB integration. They will invariably use Microsoft Windows as a framework, and a few will have Java too. Small companies are highly likely to use outside services for SOA implementations, especially via local VARs. When they choose, small companies will single-source SOA and ESB technology from a single vendor.



Figure 9 – Challenges and Responses by Company Size



Source: [AberdeenGroup](#), July 2006

Medium Companies with \$50M to \$1B in Revenues

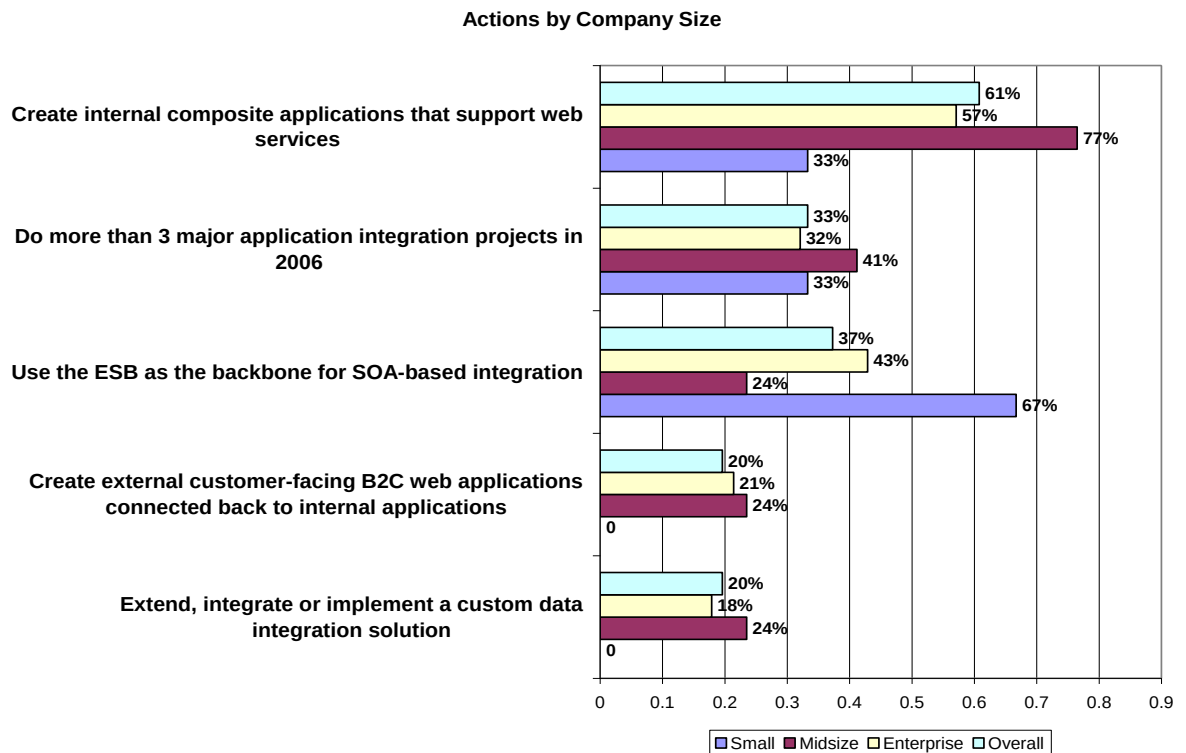
A quarter of mid-size companies will get started in SOA planning and design this year, and less than a fifth have more than a year's experience. Figure 9 shows a variety of challenges, with creating composite applications with Web services being the most frequent response to the challenges. We expect *SOA Lite* to be a common model among mid-size companies, as will *SOA ERP* due to the time, talent, and budget constraints at many mid-size firms. A quarter of mid-size firms will do B2B applications this year, and 30% will do B2C. About 40% say they will do more than 3 projects this year — and recall that a quarter are just getting started in SOA! Composite applications are the most frequently chosen action. The primary competitor technology to ESB was EAI. Unix, Linux, and Windows are the platforms chosen, with some mainframes. Medium-size companies are unlikely to outsource ESB implementations, but half will work along side outside SOA experts on initial implementations. A majority plan to standardize on SOA software from a single vendor.

Large Companies with Revenues Exceeding \$1B

Over 40% of large companies have more than a year's experience in SOA planning, design, and programming. They are by far the most mature group. About 5/6 have already installed ESB technology, and another 10% will do so this year. Large companies are most likely to have installed B2C, B2B, supply chain, and portal applications using

SOA technology. They are interested in controlling IT costs, speeding up IT implementation, and re-using applications. Three-fifths looked at EAI along side an ESB, and a third also looked at CORBA object brokers, IBM MQ and BPM software as technology substitutes for ESBs. Ease of integration flexibility with current and planned applications is the top technical buying criteria, with overall technical attractiveness and projected ROI/ROA being the top two other buying criteria. Large companies plan to use Unix over Linux by about 3:1, with almost half using SOA as an entrée to mainframes. A third plan more than three SOA applications this year, with over half mentioning composite application development. About 40% will use outside services for initial implementations while the same will work along side outside contractors to gain experience.

Figure 10 – Challenges and Responses by Company Size



Source: [AberdeenGroup](#), July 2006

Business Value Findings by Industry Group

Table 3 shows the business drivers of four major industrial categories. Table 4 depicts the responses to the challenges by industrial category. See Appendix A for the breakdown into the four categories.

**Table 3: Top Three Business Drivers by Industry Group (Multiple Choice)**

	Overall	Discrete/ Process	Consu- mer	Services	Public
Re-usage of applications via Web Services	37%	33%	27%	47%	33%
Managing IT integration costs	37%	40%	46%	35%	67%
Speed of IT implementations	31%	27%	36%	47%	0%
Alignment with the business	29%	53%	27%	12%	67%
Management of IT complexity	28%	27%	0%	29%	33%
Development of new business capabilities or new products and services	26%	20%	46%	18%	33%

Source: [AberdeenGroup](#), July 2006

Discrete and Process Manufacturing sectors are most focused on aligning IT with the business, followed by managing IT costs. Almost half of discrete and process manufacturers will do more than three SOA application integration projects in 2006, the largest contingent of the four sectors. Discrete/process are more likely to create complex composite applications across multiple ISV application software products, making this sector ideal for an *SOA ERP* tool kit.

Consumer Goods companies are driven by IT cost management and speed of IT implementations. This sector is most likely to drive applications for new products or business services, and least likely to be challenged by IT complexity.

Services sector is equally driven by speed of IT implementations and re-use of applications via Web services, especially via composite applications. Alignment with the business is well below the overall concern. Unlike consumer companies, the typical services company is not interested in an *SOA ERP* strategy. Consumer goods manufacturers are most likely to embark on a B2C application, as web merchandising is a mature marketing approach.

Public Sector is most likely to be cost-focused and challenged by alignment of IT with the changing business requirements. The public sector is most likely overall to embrace *SOA ERP* to integrate multiple applications and least likely to create composites of internal applications. Unfortunately for backers of e-government, our small public sector sample has no interest in B2C applications.

Table 4: Top Three Responses to Challenges by Industry Group (Multiple Choice)

	Overall	Discrete/ Process	Consu- mer	Services	Public
Create internal composite applications that support web services	61%	47%	73%	71%	33%
Use the ESB as the backbone for SOA-based integration	37%	33%	36%	41%	33%
Do more than 3 major application integration projects in 2006	33%	47%	36%	29%	33%

Enterprise Service Bus: an SOA Middleware Foundation

Span multiple vendor's enterprise applications (i.e., ERP, CRM) as composite apps	22%	33%	9%	12%	67%
Create external customer-facing B2C web applications connected back to internal applications	20%	13%	9%	35%	0%

Source: [AberdeenGroup](#), July 2006



Chapter Four: Recommendations for Action

Key Takeaways

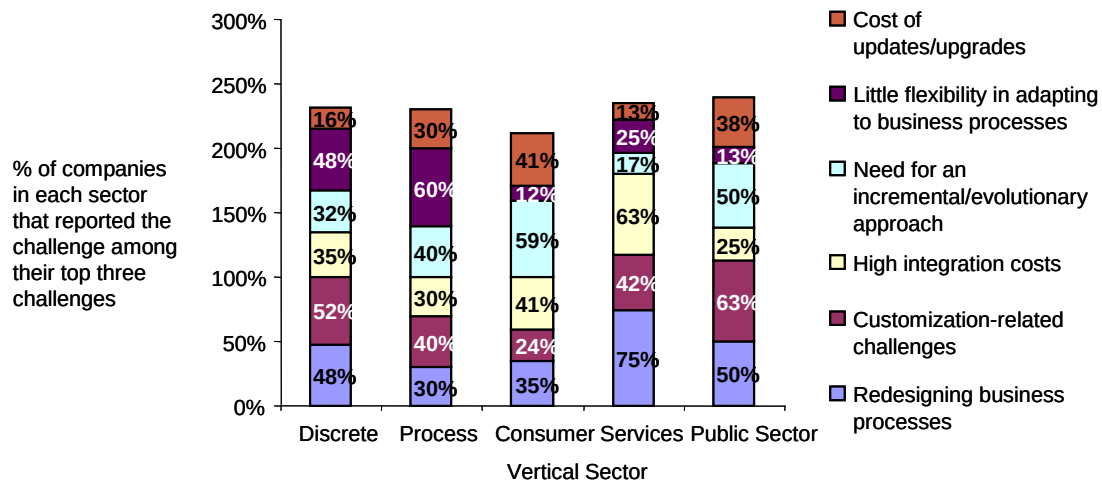
- If an ISV application has adequate functionality for your organization, plan to upgrade to the SOA-enabled version.
- SOA is neither a panacea nor a free lunch. Look at organizational discipline improvements and SOA competency training before leaping.
- Measure twice and cut once: to get to an enterprise-wide SOA infrastructure will take several years of planned but incremental growth and deployments.

Application integration is a never-ending process – it is unlikely that a thriving business will reach a point of business-process stasis. Unfortunately, too many organizations have not invested in the training and management discipline to make this on-going application integration process a well-oiled machine.

It is clear from this study that the next-generation of business applications will have deep business intelligence views into complex LOB processes, and many organizations will move towards a machine-in-loop model of business process management where humans deal with the exceptions to the rules. Both will result in better management due to transparency and immediacy. Notably, your peers believe SOA technology is the critical technology glue that will make application integration less complex, more repeatable, faster, and more flexible.

The business challenges faced by organizations vary by vertical industry sector. (Figure 6):

- The public sector is most challenged in customization and related costs. This explains the world-wide interest in e-government.
- Services are plagued by high integration costs.
- Discrete and process manufacturers are more concerned by flexibility in redesigning business processes, and customization to fit changing business processes. Process is particularly plagued with three-fifths lacking flexibility in adapting software to business processes, while half of discrete manufacturers have the same problems.
- Consumer products companies complain most about the challenges of radical changes: they want a more measured, evolutionary approach.

Figure 6: Top Challenges by Vertical Sector

Source: AberdeenGroup, April 2006

Laggard Steps to Success

1. *Evaluate the ISV application set for functional completeness.*

Do not charge off with new technology if the ISV application horse is not up to the task! There are numerous modern applications for use in-house or through an application service provider (ASP). On a five-year basis, it is likely less expensive to change application horses for a modern technology base with the right functionality than pay outside service providers to patch up an inadequate application. Consider buying the right vertical-market-specific ISV application, and outsourcing to an ASP.

2. *Install the ISV's SOA-enabled application versions. Train on this baseline.*

Start with a solid technology base, which includes the ISV application versions which incorporate SOA technology. One approach is to get the ISV's upgrade, install it, and use that SOA technology as a baseline for training and learning about SOA.

3. *Conduct an SOA readiness assessment.*

Many service companies now provide comprehensive SOA readiness assessments. Such an assessment becomes the planning roadmap for technology acquisition, application integration priorities for individual projects, and organizational readiness to stage a new integration technology roll-out.

Industry Norm Steps to Success

1. *Perform gap analysis on ISV SOA application versions.*

While every current version may be SOA-enabled, it is unlikely they will all mesh and merge seamlessly into a full-blown enterprise SOA infrastructure. First, determine how well individual applications integrate with the functionality of their SOA peers. Then, consider how these applications will integrate into a



full-blown SOA infrastructure that includes an enterprise service bus (ESB), business process management (BPM), a metadata repository, and a cross-platform development environment including middleware.

2. *Build your SOA infrastructure project by project.*

Prioritize LOB requirements for application integration based on ROI or other metrics. Map the technology requirements of the project to the overall infrastructure plan, and buy accordingly.

3. *Work on the organizational and technology maturity issues.*

Integration competency is a hallmark of best-in-class organizations, as is organizational discipline. Measure and drive down integration costs and timeframes while improving skill-sets.

4. *Catch up on the integration backlog.*

A hallmark of best-in-class organizations is the lack of significant application issues. That means the IT organization is scaled and competent at keeping the backlog manageable and completion rates timely for the LOB. The implication is that a new SOA technology toolset needs to be completely mastered.

Best in Class Next Steps

1. *Build it and they will come.*

BIC organizations are already mature in IT organization, technology selection and technology integration areas. Their opportunity is to build out an SOA infrastructure that integrates the enterprise's applications while providing an outward facing posture to customers, partners, and suppliers.

2. *The next stage is real-time BPM with machine-in-the-loop.*

Several Aberdeen studies point to the need for deeper visibility into active business processes with RPPM. First, look to BPM software with knowledge management, analytics, and rules-based exception handling. Next, work with LOB process owners to drive out human bottlenecks to efficient process flow through exception-based application-integration reporting. Use rules-based technology to make decisions and ensure compliance, removing unnecessary human intervention.

3. *Plan for the heterogeneous SOA integration environment.*

SOA is no one trick pony. SOA architecture in many larger organizations will be heterogeneous, with the need to plan for the complex, phased implementation across operating system platforms, applications, geographies, and lines of business. To do that successfully will require enterprise-level planning and governance.

Featured Sponsors



Sponsor Directory

Author Profile

Peter S. Kastner,
Research Vice President and Co-Founder
Enterprise Integration Research
AberdeenGroup, Inc.

Area of Focus

Peter S. Kastner conducts research in the integration of information technology across and among enterprises and their customers and employees. He focuses on the near-term business value-generation tradeoffs faced by technology and business managers who must deliver repeated, short-term value gains while simultaneously refreshing the enterprise infrastructure to take advantage of evolving technology, standards, practices, and regulations. The practice covers technology integration including application development; software architectures and middleware; core architecture, processing and networking; edge devices, processing, and networking; data and business intelligence and integration; security integration; managed services and integration third-party services; and, enterprise business process and technology management.

Current Research

Kastner is delivering a research agenda centered on determining the 2006 technology tradeoffs made by business managers, the plans and experiences of enterprises with application-oriented networks, next-generation approaches to ERP integration, mid-market systems-oriented architecture (SOA) decisions, and best practices in SOA technology. Second-half research will benchmark the numerous technologies and approaches to enterprise integration, probe buyer experiences with next-generation datacenter technologies, and research business process management and real-time operational decision-making technologies and buyer experiences with next-generation data warehouses.

Experience

Kastner's career includes a decade in systems integration, operations, and application development as a technician and manager; a decade in computer company product marketing, vertical market development, sales management, and training; and 18 years of market research experience in software and hardware technologies at Aberdeen Group. He has been cited hundreds of times in the press, has appeared on all the major U.S. news and business networks, and is a frequent speaker on issues about the business and consumer applications of technology. He is a founder of the Transaction Processing Council. In 2003, he was named one of the top five most sought after analysts by MediaMap.



Appendix A: Research Methodology

Between January and March 2006, **AberdeenGroup** examined the application, integration competency, experiences, and intentions of enterprises in discrete manufacturing, process manufacturing, consumer product, service, and government industries.

Responding IT and LOB technology executives completed an online survey that included questions designed to determine the following:

- The degree to which application integration issues impact corporate and IT strategies, operations, and financial results
- The structure and effectiveness of existing application integration management procedures
- Current and planned use of technology to improve application integration and the ROI of ISV application investments
- How enterprises are extracting more value from their enterprise applications (especially ERP's and legacy custom applications)
- Will it enable flexible business process integration, manipulation, orchestration better based on more flexible workflows?

Aberdeen supplemented this online survey effort with telephone interviews with select survey respondents, gathering additional information on enterprise applications management strategies, experiences, and results.

The study aimed to identify emerging best practices for application integration and provide a framework by which readers could assess their own enterprise application integration capabilities.

Responding enterprises included the following:

- **Job title/function:** The research sample included respondents with the following job titles:

Senior Management (CEO, CFO, COO)	11%
CIO/IT Leader	15%
(Senior) Vice President	6%
Director	19%
Manager	25%
Staff	6%
Internal Consultant	11%

Other 6%

- **Industry:** The research sample included respondents predominantly from manufacturing industries. Automotive manufacturers represented 22% of the sample, followed closely by high-tech manufacturers, which accounted for 20% of respondents. Manufacturers of metals and metal products (including industrial equipment) totaled 16% of respondents. Aerospace and defense manufacturers accounted for 10% of the sample. Other sectors responding included medical equipment, construction/engineering, and retail and distribution.

- **Geography:**

North America (Includes USA, Canada, Mexico)	55%
Europe	23%
Asia/Pacific	8%
Middle East, Africa	7%
South/Central America and Caribbean	6%

- **Company size:**

Less than \$50 Million	31%
\$50M to \$249M	20%
\$250M to 499M	17%
\$500M to \$999M	10%
\$1 billion to \$5 billion	17%
More than \$5 billion	4%

Solution providers recognized as sponsors of this report were solicited after the fact and had no substantive influence on the direction of the *Achieving More Value from Enterprise Applications Benchmark Report*. Their sponsorship has made it possible for [AberdeenGroup](#) to make these findings available to readers at no charge.



Appendix B: **Related Aberdeen Research & Tools**

Related Aberdeen research that forms a companion or reference to this report includes:

- The Service-Oriented Architecture (SOA) in IT Benchmark Report (December 2005)
- The Service-Oriented Architecture in the Supply Chain Benchmark Report (September 2005)
- The Strategic Enterprise IT Budget Realities
- The 2005 CIO Agenda — CIO Disruptors
- The IT Spend Benchmark Report
- SOA Success Starts with IT Success
- The ESB in the Land of SOA
- “Supply Chain as a Service”: The Next Big Thing?
- Creating Composite Business Process Applications for Enterprise Integration
- Sector-Based Research
 - For Mid-Size Enterprises, SOA’s Benefits Begin with IT
 - Process Manufacturers Cautious on SOA Adoption
 - SOA is Made to Order for Mid-Size Manufacturers
 - Complexity Drives Discrete Manufacturers to SOA
- Key Performance Indicators
 - Software Maintenance Costs as a Percent of IT Budget
 - IT Cost of Delivering/Building New Applications as a Percent of IT Budget
 - Percent of IT Budget Spent on Innovations

Information on these and any other Aberdeen publications can be found at www.Aberdeen.com.

About **AberdeenGroup**

Our Mission

To be the trusted advisor and business value research destination of choice for the Global Business Executive.

Our Approach

Aberdeen delivers unbiased, primary research that helps enterprises derive tangible business value from technology-enabled solutions. Through continuous benchmarking and analysis of value chain practices, Aberdeen offers a unique mix of research, tools, and services to help Global Business Executives accomplish the following:

- IMPROVE the financial and competitive position of their business now
- PRIORITIZE operational improvement areas to drive immediate, tangible value to their business
- LEVERAGE information technology for tangible business value.

Aberdeen also offers selected solution providers fact-based tools and services to empower and equip them to accomplish the following:

- CREATE DEMAND, by reaching the right level of executives in companies where their solutions can deliver differentiated results
- ACCELERATE SALES, by accessing executive decision-makers who need a solution and arming the sales team with fact-based differentiation around business impact
- EXPAND CUSTOMERS, by fortifying their value proposition with independent fact-based research and demonstrating installed base proof points

Our History of Integrity

Aberdeen was founded in 1988 to conduct fact-based, unbiased research that delivers tangible value to executives trying to advance their businesses with technology-enabled solutions.

Aberdeen's integrity has always been and always will be beyond reproach. We provide independent research and analysis of the dynamics underlying specific technology-enabled business strategies, market trends, and technology solutions. While some reports or portions of reports may be underwritten by corporate sponsors, Aberdeen's research findings are never influenced by any of these sponsors.



↓ ↓ DO NOT DELETE SECTION BREAK BELOW THIS LINE ↓ ↓
It should always be on the bottom of the second-to-last page,
so the boilerplate will appear in the last page footer

AberdeenGroup, Inc.
260 Franklin Street, Suite 200
Boston, Massachusetts
02110-3112
USA

Telephone: 617 723 7890
Fax: 617 723 7897
www.aberdeen.com

© 2006 **AberdeenGroup, Inc.**
All rights reserved
April 2006

Founded in 1988, **AberdeenGroup** is the technology-driven research destination of choice for the global business executive. **AberdeenGroup** has over 100,000 research members in over 36 countries around the world that both participate in and direct the most comprehensive technology-driven value chain research in the market. Through its continued fact-based research, benchmarking, and actionable analysis, **AberdeenGroup** offers global business and technology executives a unique mix of actionable research, KPIs, tools, and services.

The information contained in this publication has been obtained from sources Aberdeen believes to be reliable, but is not guaranteed by Aberdeen. Aberdeen publications reflect the analyst's judgment at the time and are subject to change without notice.

The trademarks and registered trademarks of the corporations mentioned in this publication are the property of their respective holders.